



Cambridge International AS & A Level

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FURTHER MATHEMATICS

9231/14

Paper 1 Further Pure Mathematics 1

October/November 2025

2 hours

You must answer on the question paper.

You will need: List of formulae (MF19)

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined page at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.

INFORMATION

- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [].

This document has **20** pages. Any blank pages are indicated.



1 Prove by mathematical induction that $\left(\frac{6}{5}\right)^n \geq 1 + \frac{1}{5}n$ for all positive integers n . [5]

[illegible]



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[illegible]



- 2** The cubic equation $x^3 + bx^2 + cx + d = 0$, where $d \neq 0$, has roots α, β, γ such that $\gamma = \frac{1}{\beta}$.

(a) Show that $\beta + \frac{1}{\beta} = d - b$.

[3]

[illegible]

(b) Show also that $\beta + \frac{1}{\beta} = \frac{1-c}{d}$.

[2]

[illegible]



(c) It is given that $b = 3$, $c = -3$ and $d > 0$.

(i) Find the value of d .

[2]

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(ii) Find the value of $\alpha^2 + \beta^2 + \gamma^2$.

[2]

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3 (a) Use standard results from the list of formulae (MF19) to show that

$$\sum_{r=1}^n (r+1)(r+2)(r+3) = \frac{1}{4}n(n^3 + bn^2 + cn + d),$$

where b , c and d are integers to be determined. [3]

This image shows a full page of white paper with horizontal dotted lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the page.



- (b) Express $\frac{2}{(r+1)(r+2)(r+3)}$ in partial fractions and hence use the method of differences to find

$$\sum_{r=1}^n \frac{2}{(r+1)(r+2)(r+3)}. \quad [5]$$

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- (c) State the value of $\sum_{r=1}^{\infty} \frac{2}{(r+1)(r+2)(r+3)}$. [1]

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4 The points A, B, C have position vectors

$$3\mathbf{i} + 5\mathbf{j} + 5\mathbf{k}, \quad 2\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}, \quad 2\mathbf{i} + \mathbf{j} - \mathbf{k},$$

respectively, relative to the origin O .

(a) Find the equation of the plane ABC , giving your answer in the form $ax + by + cz = d$. [5]

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(b) The point D has position vector $\mathbf{i} - \mathbf{j} + 3\mathbf{k}$.

Find the shortest distance between the lines AB and CD .

[5]





5 (a) (i) By writing $\cos\left(\frac{1}{12}\pi\right)$ as $\cos\left(\frac{1}{4}\pi - \frac{1}{6}\pi\right)$, show that $\cos\left(\frac{1}{12}\pi\right) = \frac{1}{4}(\sqrt{6} + \sqrt{2})$. [1]

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(ii) Show also that $\sin\left(\frac{1}{12}\pi\right) = \frac{1}{4}(\sqrt{6} - \sqrt{2})$. [1]

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The matrix $\mathbf{M}$ is such that $\mathbf{M} = \frac{1}{4} \begin{pmatrix} \sqrt{6} + \sqrt{2} & \sqrt{2} - \sqrt{6} \\ \sqrt{6} - \sqrt{2} & \sqrt{6} + \sqrt{2} \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$.

(b) The matrix $\mathbf{M}$ represents a sequence of two geometrical transformations in the x - y plane.

Give full details of each transformation, and make clear the order in which they are applied. [4]

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(c) Write $\mathbf{M}^{-1}$ as the product of two matrices, neither of which is $\mathbf{I}$.

[2]

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(d) Given that $y = mx$ is an invariant line of the transformation represented by $\mathbf{M}$, show that

$$m^2 \sin\left(\frac{1}{12}\pi\right) + 2m \cos\left(\frac{1}{12}\pi\right) - \sin\left(\frac{1}{12}\pi\right) = 0$$

and find the values of m in the form $a \cot\left(\frac{1}{12}\pi\right) + b \operatorname{cosec}\left(\frac{1}{12}\pi\right)$, where a and b are integers to be determined. [6]

[6]

[illegible]



- 6** The curve C has polar equation $r = \cos \frac{1}{2}\theta$, for $0 \leq \theta \leq \pi$.
- (a)** Sketch C .

[2]

In parts **(b)** and **(c)** you may use the identities $\sin \theta \equiv 2 \sin \frac{1}{2} \theta \cos \frac{1}{2} \theta$ and $\cos \theta \equiv 2 \cos^2 \frac{1}{2} \theta - 1$.

- (b) Find the exact value of the area of the region enclosed by C and the initial line. [4]

[illegible]



- (c) Find the maximum distance of a point on C from the initial line. Give your answer in the form $p\sqrt{q}$, where p and q are rational numbers to be determined. [6]

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7 The curve C has equation $y = \frac{10x^2 - 11x - 18}{10x - 18}$.

(a) Find the equations of the asymptotes of C . [3]

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(b) Show that C has no stationary points. [4]

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(c) Sketch C , stating the coordinates of the points of intersection with the axes.

[3]

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(d) Sketch the curve with equation $y = \left| \frac{10x^2 - 11x - 18}{10x - 18} \right|$.

[2]





- (e) Given that $\left| \frac{10x^2 - 11x - 18}{10x - 18} \right| < 4$ for $\frac{-29 - \sqrt{4441}}{20} < x < p$ and $\frac{-29 + \sqrt{4441}}{20} < x < q$, find the values of p and q . [4]

[illegible]



Additional page

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